SAI KUMAR

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PROFESSIONAL SUMMARY

Robotics Software Engineer with around 4 years of experience building autonomy, perception, and robot learning systems in C++ and Python. Delivered ROS2 navigation stacks, RL locomotion and VLA manipulation policies in Isaac Sim and Isaac Lab, and TensorRT-optimized perception on Jetson. Background in embedded control (CAN Bus, FreeRTOS, ARM Cortex) and IEC 61508-aligned CI/CD, with a focus on sim-to-real transfer and edge deployment.

SKILLS

Programming Languages: C++, Python, CUDA, MATLAB

Robotics Frameworks: ROS2, Nav2, MoveIt2, ros2_control, BehaviorTree.CPP, Micro-ROS, DDS/QoS tuning

Autonomy & Motion Planning: SLAM (LiDAR & Visual), MPC, RRT*, OMPL, Multi-Robot Coordination

Embodied AI & Robot Learning: Vision-Language-Action (VLA) Models, Imitation Learning & Diffusion Policies, RL for Locomotion & Whole-Body Control, LLM-Based Task Planning

Perception & AI: PyTorch, OpenCV, YOLO, Sensor Fusion, 3D Mapping, Visual-Inertial Odometry

Simulation & Sim-to-Real: Isaac Sim, Isaac Lab, Gazebo, HIL/SIL Testing

Embedded & Control: Embedded Linux, FreeRTOS, ARM Cortex, Motor Control, CAN Bus, EtherCAT

Deployment & Infrastructure: Docker, GitHub Actions, GitLab CI/CD, ONNX Runtime, TensorRT, Edge AI Deployment

Safety & Standards: ISO 10218, ISO/TS 15066, IEC 61508

WORK EXPERIENCE

Robotics Software Engineer, NVIDIA, United States **September 2025 - Present**

- Trained reinforcement learning locomotion policies in Isaac Lab across 4,096 parallel GPU environments with domain randomization, raising legged-robot sim-to-real success from 58% to 89% and cutting iteration time from 3 days to 9 hours.
- Optimized PyTorch YOLO and visual-inertial odometry models with TensorRT and ONNX Runtime for Jetson Orin, reducing end-to-end perception latency from 48 ms to 19 ms (60%) and enabling 30 FPS onboard inference.
- Engineered a ROS2 autonomous navigation stack using Nav2, LiDAR and visual SLAM, and 3D mapping, tuning DDS QoS profiles to cut message loss from 7% to under 1% and lowering localization drift by 35% in multi-room tests.
- Fine-tuned Vision-Language-Action and diffusion policy models on 1,200+ teleoperated demonstrations in Isaac Sim, integrating with MoveIt2 and ros2_control to improve manipulation task from 62% to 84% across pick-and-place scenarios.
- Automated software-in-the-loop and hardware-in-the-loop regression testing with Docker, GitHub Actions, and Isaac Sim, shortening release validation and catching 40+ integration defects before deployment.

Associate Robotics Software Engineer | Kpit Technologies, India **March 2022 – August 2024**

- Developed a C++ Model Predictive Control (MPC) path-tracking controller for an autonomous vehicle platform, validated in MATLAB, cutting lateral tracking error from 0.45 m to 0.18 m at speeds up to 30 km/h.
- Implemented CAN Bus communication and FreeRTOS tasks on ARM Cortex-M controllers for motor control and sensor data, achieving zero dropped frames over 100+ hours of bench testing and a 99.9% on-time message delivery rate.
- Built a LiDAR and camera sensor fusion pipeline with OpenCV and 3D mapping, improving obstacle detection recall from 81% to 94% and reducing false positives by 28% across 50+ recorded drive scenarios.
- Designed RRT* motion planners with OMPL and tested them in Gazebo, cutting planning time by 40% and reaching a 96% collision-free success rate spanning 500 simulated scenarios.
- Integrated Micro-ROS on embedded controllers with an Embedded Linux gateway, reducing boot-to-ready time from 45 s to 18 s and cutting custom firmware code by 30%.
- Established a GitLab CI/CD pipeline with automated static analysis and unit tests aligned to IEC 61508 safety requirements, covering 85% of safety-critical modules and catching defects before integration throughout 20+ releases.

PROJECTS

Agentic Data Science Copilot | *Python, Llama 3, LangGraph, LangChain, SQL, Pandas, DuckDB, PostgreSQL, Gradio*

- Designed an agentic data-analysis copilot using Llama 3 and LangGraph to translate natural-language questions into SQL and Python workflows, coordinating schema discovery, query generation, validation, execution, and visualization.
- Configured tool calling, structured outputs, conversation memory, human-approval checkpoints, and evaluation for SQL correctness, task completion, and response quality to improve reliability of generated analytics.

Multimodal Document Intelligence | *Python, Llama 3, Hugging Face, LayoutLMv3, OCR, OpenCV, PyMuPDF, Streamlit*

- Created a multimodal document-intelligence pipeline using OCR, LayoutLMv3, OpenCV, and transformer models to classify scanned forms and extract structured fields from invoices and business documents.
- Validated low-confidence extractions using confidence scoring, validation rules, and human-in-the-loop review, measuring field-level precision, recall, and F1-score across varied document layouts.

EDUCATION

Master of Science in Information Systems | Northeastern University, Boston, MA